

CSE 309: Theory of Automata

Reducibility

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April 2024

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Reducibility

Reduction

A **reduction** is a way of converting one problem to another problem in such a way that a solution to the second problem can be used to solve the first problem. Such reducibilities come up often in everyday life, even if we don't usually refer to them in this way.

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- E.g., suppose that you want to find your way around a new city
- You know that doing so would be easy if you had a map
- Thus, you can reduce the problem of finding your way around the city to the problem of obtaining a map of the city

Reduction

- Reduction involves two problems (we call them A and B)
- If A reduced to B , we can use a solution to B to solve A
- So, in our example, A is the problem of finding your way around the city, and
- B is the problem of obtaining a map

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- If A reduced to B , we can use a solution to B to solve A
- So, in our example, A is the problem of finding your way around the city, and
- B is the problem of obtaining a map
- **Important:** reduction does not say anything about solving A and B alone but only about the solvability of A in the presence of a solution to B

Undecidability Problems from Language Theory

- Consider the problem $HALT_{TM}$, the problem of determining whether a Turing machine halts (by accepting or rejecting) on a given input
- We will use undecidability of A_{TM} to prove the undecidability of the halting problem by reducing A_{TM} to $HALT_{TM}$

$$HALT_{TM} = \{\langle M, w \rangle : M \text{ is a TM and } M \text{ halts on input } w\}$$

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Theorem 1

$HALT_{TM}$ is undecidable.

Proof for Undecidability of $HALT_{TM}$

- Proof is by contradiction
- Assume $HALT_{TM}$ is decidable i.e., some TM R exists that decides $HALT_{TM}$
- We construct TM S to decide A_{TM} with S operating as follows:
 $S =$ “On input $\langle M, w \rangle$, an encoding of TM M and a string w
 1. Run TM R on input $\langle M, w \rangle$
 2. If R rejects, *reject*
 3. If R accepts, simulate M on w until it halts
 4. If M has accepted, *accept*; if M has rejected, *reject*.”
- If R decides $HALT_{TM}$, then S decides A_{TM}
- Since A_{TM} is undecidable, $HALT_{TM}$ also must be undecidable.

Undecidability Problems from Language Theory

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- Now we consider the language EQ_{TM} ,

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Theorem 4

EQ_{TM} is undecidable.